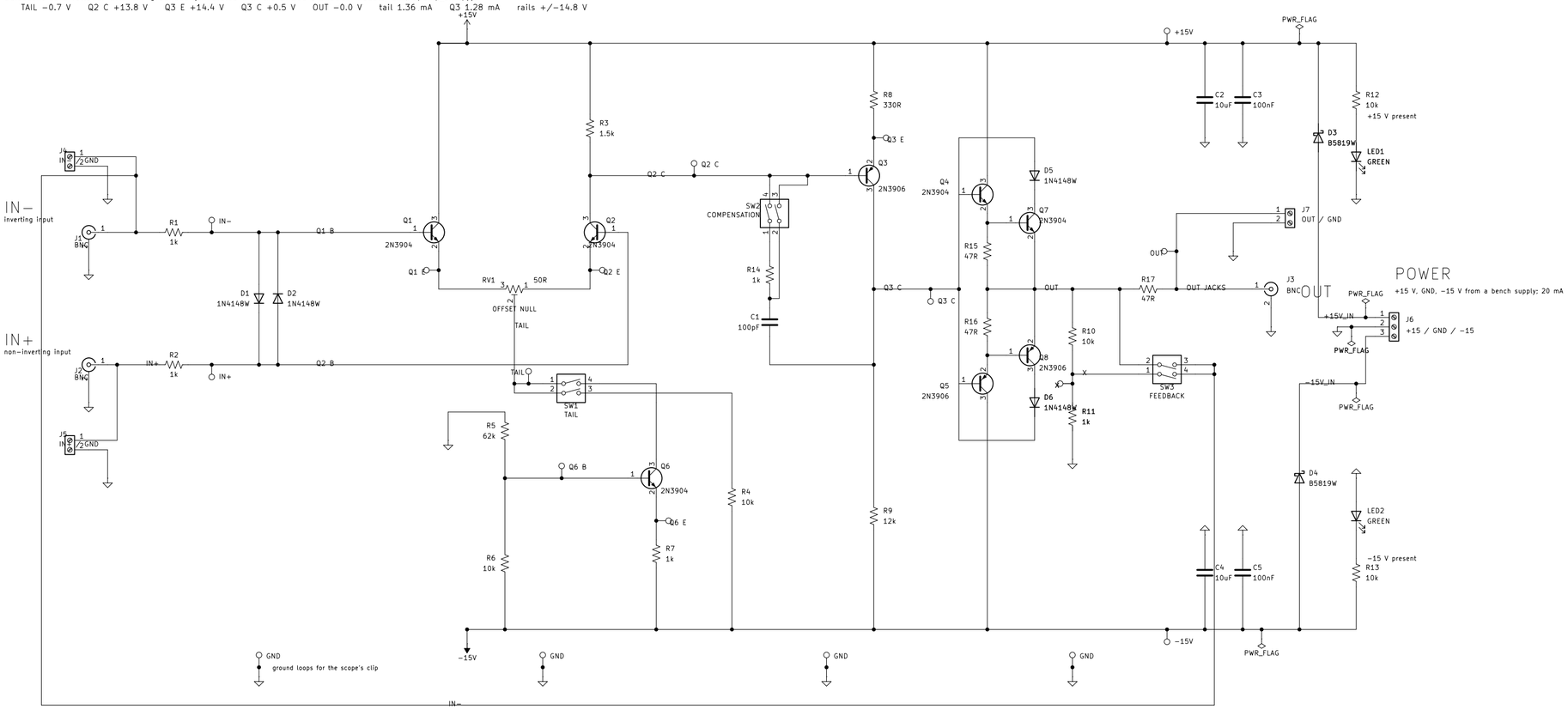
DISCRETE TRANSISTOR OP-AMP

The op-amp of Lab 5L (Learning the Art of Electronics, Hayes & Horowitz, 2016), built from the NPN and
PNP of Chapter 4 as eight SOT-23 transistors, on a board whose silkscreen is this drawing.
Three stages, as the book's Fig. 5L.11 and 5N.32:

difference	Q1 Q2, 1.5k collector load on the + side, tail from RV1 through SW1	x11
gain	Q3, PNP common emitter: 12k / (330 + re) = 32; the resting push-pull's dead zone leaves a third of it	x11
follower	Q4 Q5 push-pull, unbiased, so crossover distortion shows until feedback hides it	x1

Open-loop gain at DC about 126 (the book says 'approximately 1000' for its CA3096 version, before the
dead zone). As a follower it settles in 0.6 us with 54 deg of phase margin; the x11 amplifier
has a bandwidth of 398 kHz and a gain of 10.1, 8 % low, as the book predicts for so little loop gain.

Quiescent as a follower with IN+ grounded (simulated with the 2N3904/2N3906 models, sim/opamp_sim.py):
TAIL -0.7 V Q2 C +13.8 V Q3 E +14.4 V Q3 C +0.5 V OUT -0.0 V tail 1.36 mA Q3 1.28 mA rails +/-14.8 V



INPUTS
Each input has a BNC and a screw terminal in
parallel; +/-12 V on either. For a common-mode
test, tee one generator into both jacks.

INPUT PROTECTION (R1, R2, D1, D2: not in the book)
1k in series with each input, and two 1N4148
back to back between the bases: the bases never
differ by more than 0.7 V, so no base-emitter
junction sees its 6 V reverse breakdown when one
input is driven hard and the other is not.
A 1 M probe or a 50 ohm generator never notices
the 1k; Ib (about 2 uA) drops 2 mV across it.

TAIL (SW1)
1 ON: Q6, the 'simple sink' of Fig. 5L.10, 1.4 mA
2 ON: 10k to −15 V, 1.4 mA (Fig. 5L.1)
The sink raises the tail's impedance from 10k to
about 1 M; stage 1's common-mode gain falls from
0.076 to 0.0004. Both OFF: no tail current, dead.
Normal: 1 ON, 2 OFF.

OFFSET NULL (RV1: not in the book)
The two halves of the 50 ohm trimmer are the pair's
emitter resistors, 25 ohm each at centre. Turning it
unbalances the pair by up to +/-33 mV at the input:
enough to centre the output in open loop and to
cancel the pair's Vbe mismatch in closed loop.
It costs half of stage 1's gain (11 instead of 20).

COMPENSATION (SW2)
1 ON: 100 pF from Q3's collector to its base, Miller-
multiplied by the stage's gain of 30 into a 3 nF
low-pass at Q2 C. The book's Fig. 5L.17.
2 ON: the bare 100 pF, no 1k: the book's version.
1 and 2 ON: the same as 2.

The 1k is not in the book. A bare Miller cap feeds
the signal FORWARD above 4 MHz, the loop's sign
flips, and a follower oscillates: the book's 'risky
business'. With the 1k the follower has 54 deg of
phase margin and settles in 0.6 us; the x11 amp is
stable either way. Both OFF: no compensation,
and the x11 amplifier rings.
Normal: 1 ON, 2 OFF.

FEEDBACK (SW3)
1 ON: X to IN-, gain 11 (Fig. 5L.16)
2 ON: OUT to IN-, a follower (Fig. 5L.18)
Both OFF: nothing; wire your own network between
the OUT, IN- and GND terminals. The feedback goes
to the IN- terminal, so R1 and the clamp still guard
Q1's base. The 10k + 1k are the book's load and its
x11 divider in one: X = OUT / 11.

OUTPUT
-14 to +13 V into the 11k; about +/-12 V, 15 mA into 1k.
Q7, Q8, D5, D6 and R15, R16 are not in the book: they
limit the output to about 17 mA, as a real op-amp does:
a short to ground is survived, to a rail for seconds.
R17 (not in the book either) feeds the jacks from OUT:
the feedback is taken before it, so a cable's 100 pF
stays outside the loop instead of making it ring.

POWER
+15 V, GND, −15 V from a bench supply, 20 mA; +/-16 V max.
Series Schottkys: a supply connected backwards
does nothing instead of forward-biasing every
transistor. The rails run 0.17 V below the
terminals: +/-14.8 V on the board. Everything
in the book's circuit is referred to those rails,
and nothing in it minds. A green LED per rail
says the supply is on and the right way round.

PROBING
A wire loop on every node named on the sheet;
a scope hook grabs it. Four ground loops for
the clip.



M3 mounting holes, one per corner

Open source, CERN-OHL-P-2.0 -- github.com/gallicchio/op-amp-pcb
difference x11, gain stage x32, follower x1 (a third at rest, in its dead zone): A = 126 open loop
2N3904 / 2N3906 (MMBT, SOT-23): the transistors of Chapter 4
Circuit: Hayes & Horowitz, Learning the Art of Electronics (2016), Lab 5L | Board: Jason Gallicchio
Sheet: /
File: opamp.kicad_sch

Title: Discrete Transistor Op-Amp: the Lab 5L op-amp of Learning the Art of

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